



## Lecture 2

# User-centric approaches

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# Goals of this lecture

1

**Distinguish** between platform- and user-centric approaches to digital trace data collection.

2

**Understand** how commonly used user-centric approaches work and how they can be used for research.

3

**Evaluate** the advantages, challenges and ethical considerations of user-centric approaches.

4

**Apply** by learning from example studies that show how research questions can be answered with user-centric approaches.

# Literature

- **WAS chapter 1:** Wearables, apps and sensors as novel tools for data collection
- **Boeschoten et al. (2022):** A framework for privacy preserving digital trace data collection through data donation
- **Ram et al. (2019): Screenomics:** A new approach for observing and studying individuals' digital lives
- **Hendrickx (2025):** News #foryou on TikTok: A digital methods-based study

# Outline

1

**Recap:** User- versus platform-centric approaches. *Why a user-centric approach?*

2

**Discuss:** three common user-centric approaches with examples

3

**Evaluate:** the methods in light of advantages, challenges and ethical considerations

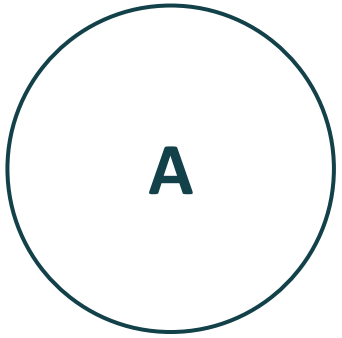
4

**Consider:** The new study designs in the context of found and designed data

01

# Recap

*User-centric versus platform-centric approaches*



## Definitions

# Digital Traces

- Also referred to as “Digital Footprint” or “Digital Shadow”
- Definition: *“One's unique set of digital activities, actions, and communications that leave a data trace on the internet or on a computer or other digital device and can identify the particular user or device.”*

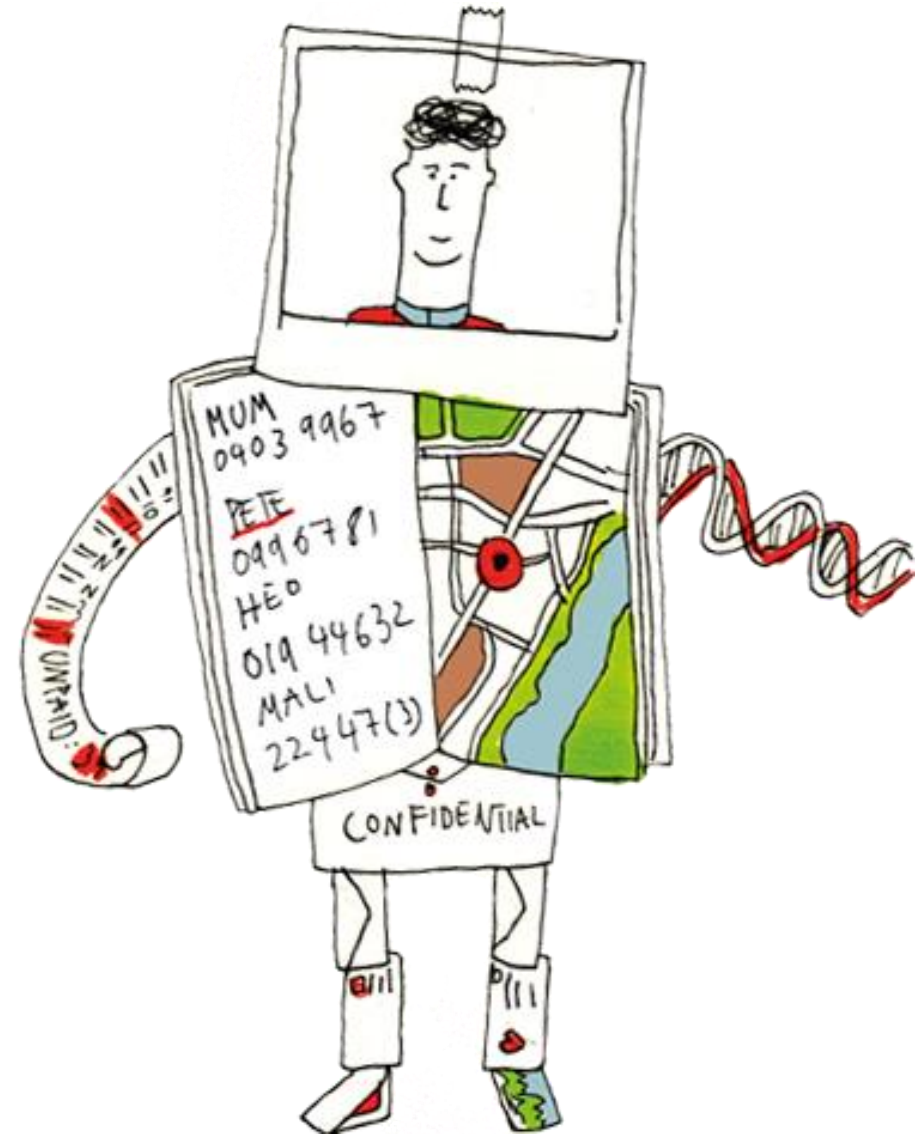
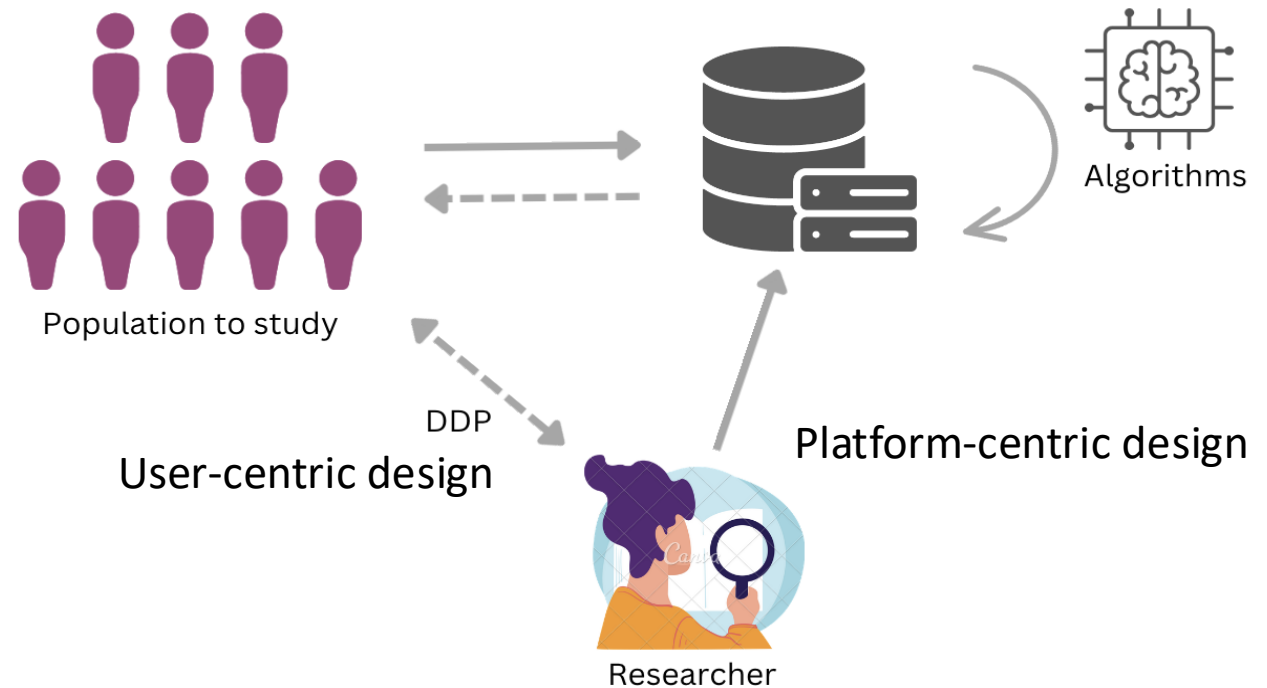


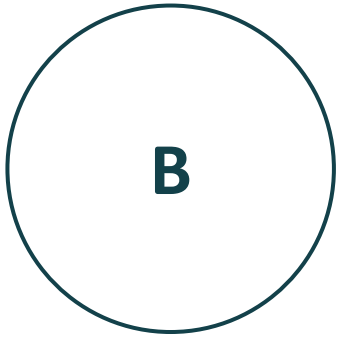
Image: <https://myshadow.org/tracking-data-traces>

Definition: <https://www.dictionary.com/browse/digital-footprint>

# User-centric and platform-centric

1. You are interested in making inferences about a population.
2. You are interested in concepts that can be measured through digital traces.
3. Platform centric → get the traces from the platform directly
4. User centric → collaborate with a sample of that population directly to obtain access to their digital traces





## **Reasons to take a user-centric approach**

# Reasons to take a user-centric approach

## New data types

- Allows for person-specific analysis (Valkenburg et al., 2021)
- Allows to capture private / non-public data (Ohme et al., 2023)

## Relation to platforms

- Allows researchers to overcome restrictions on APIs and scraping (Bruns, 2019; Breuer et al., 2023)
- Avoids platform dependency (Ohme et al., 2023)

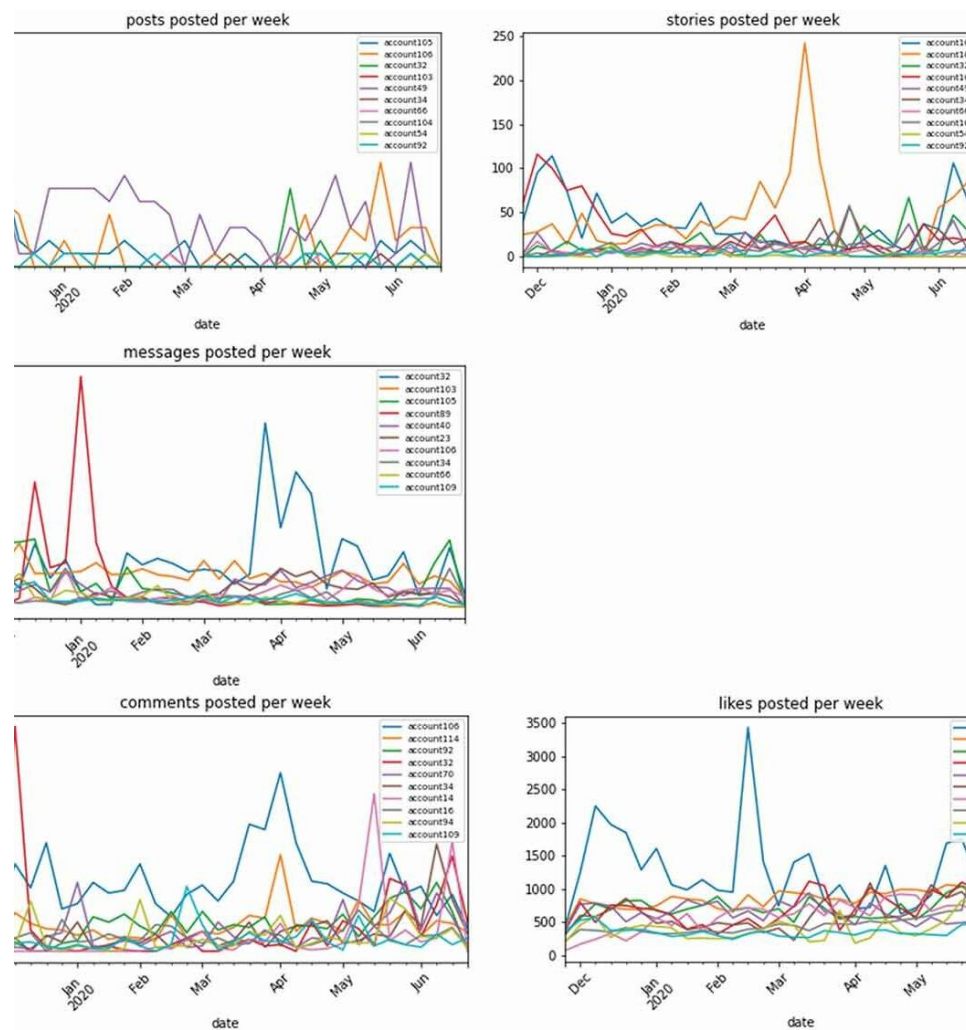
## Role of participants

- Allows to obtain active informed consent (de León et al., 2026)
- Allows to define your population and recruit your sample yourself (Breuer et al., 2023; Ohme et al., 2023)

## Data quality

- Allows to combine digital traces with other data sources on person level (Ohme et al., 2023)
- Reduces the reliance on self-reports (de León et al., 2026)

# Illustrated by Project AweSome



## Measurement Burst Design (Nesselrode, 1991)

N = 388 (age 13-15)

T > 250

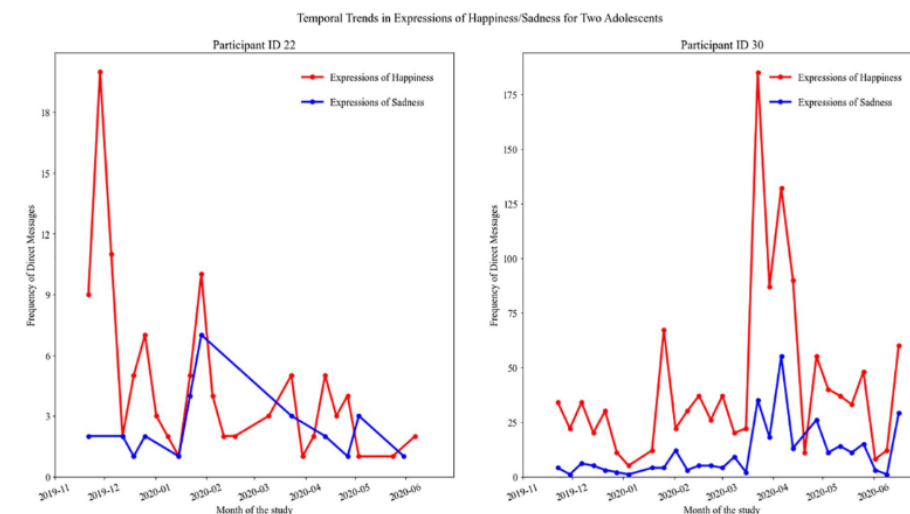


Figure 4. Temporal trends in the frequency of direct messages containing expressions of happiness (red line) and sadness (blue line) for two different participants.

Note. The y-axis differs per plot.



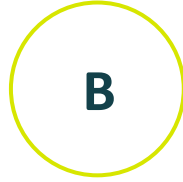
02

## Three common user-centric approaches

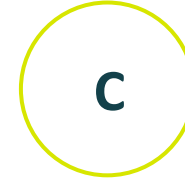
# Three common user-centric approaches



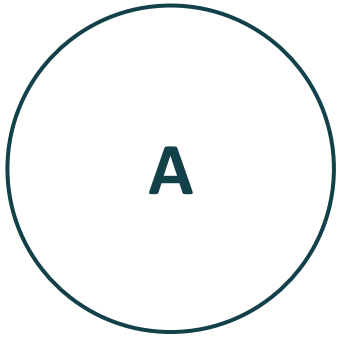
**Sensor approaches**



**Capture and scraping approaches**



**Data donation approaches**



## **Sensor approaches**

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# Sensor approaches - definitions

## DEFINITION

### Sensor

A piece of hardware that converts a physical phenomenon into an electric signal.

For example: the accelerometer sensor measures the rate of change of velocity of an object

## DEFINITION

### Wearable

A lightweight device including one or multiple sensors that people can wear close to or on their skin.

For example: a fitness bracelet or smartwatch.

## DEFINITION

### App

A software program that runs on a smartphone or other smart device. Apps allow users to interact with a device.

For researchers, apps provide the opportunity to collect, aggregate and process data collected on the device.

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# Sensor approaches – more definitions

## DEFINITION

### Passive data collection

Data are collected, for example by sensors, as a byproduct of an everyday activity.

For example: an accelerometer collects data while the participant walks.

## DEFINITION

### Active data collection

A participant deliberately creates data as a reaction to a stimulus designed by the researcher.

For example: the participant responds to a survey question to take a picture and share it with the researcher.

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Exercise

## How does it work in practice? *Choose between these alternatives*

A

### Direct sensor use

You ask participants to use the sensor directly

B

### Custom software

You develop a piece of software (e.g. app or plug-in) that captures the traces you are interested in

C

### Use existing software

You make use of an existing app or plug-in

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Exercise

## How does it work in practice? *Procedure*

1

**Invite participants to your study**

2

**Instruct them to use [A, B or C from  
the previous slide]**

3

**From that moment, data collection  
happens prospective**

*This means: Digital traces are collected as they are  
being produced.*

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# What sensors can you use?

NFC

Thermometer

Wi-Fi

Cellular network

Bluetooth

Air humidity sensor

Light sensor

Camera

Proximity sensor

Microphone

Fingerprint sensor

Compass

Barometer

Accelerometer

Pedometer

Gyroscope

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Exercise

## What can you measure with these sensors?

NFC

Thermometer

Wi-Fi

**Location**

Cellular network

Bluetooth

Air humidity sensor

Light sensor

**Ambience**

Camera

**Proximity**

Proximity sensor

Microphone

Fingerprint sensor

Compass

Barometer

Accelerometer

Pedometer

Gyroscope

**Physical activity**

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# Sensor categories

## Location

- GPS
- Cellular network
- Wi-Fi
- Beacons

## Physical activity

- Accelerometer
- Gyroscope
- Barometer
- Pedometer

## Ambience & proximity

- Camera
- Microphone
- Light sensor
- Bluetooth
- NFC

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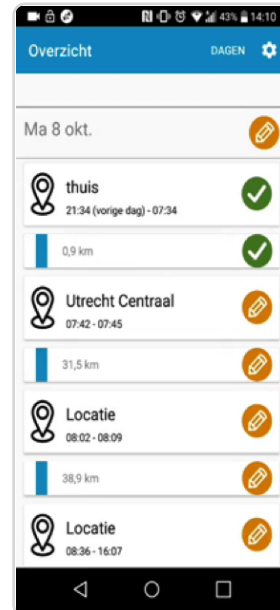
4

Exercise

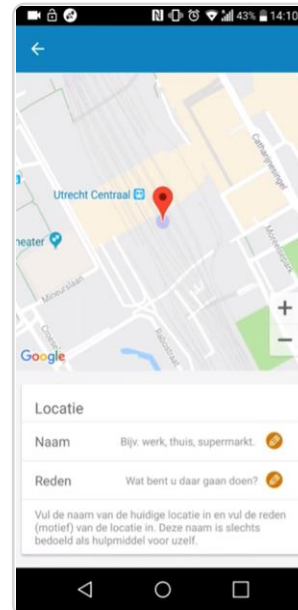
## EXAMPLE

# Measuring everyday mobility using an app

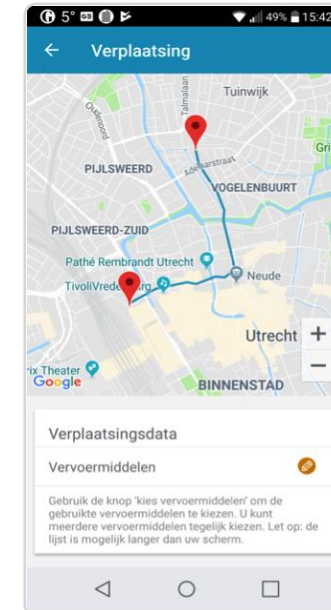
N = 1,902 · Dutch general population, field test Nov–Dec 2018: Statistics Netherlands' TABI app (Android & iOS), 7 days, GPS + Wi-Fi



Daily overview



Questions about stops



Questions about trips

1

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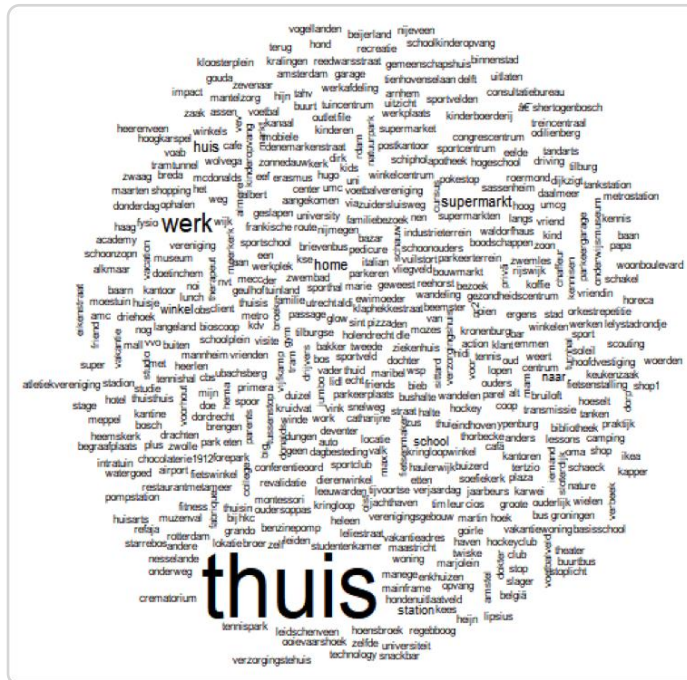
Example

4

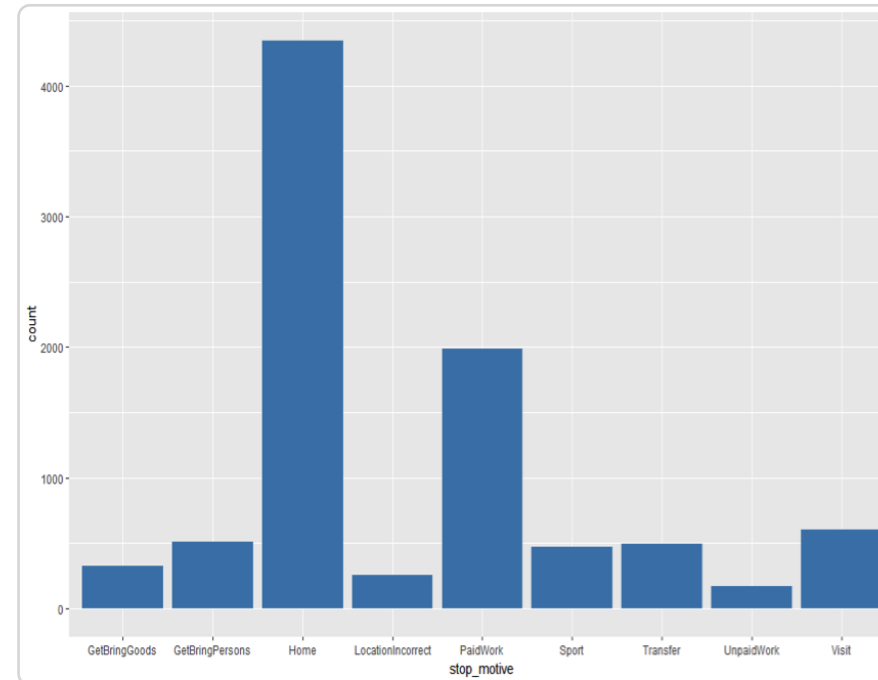
Exercise

# Motives and means of transportation

**22,000 stops logged** · about **50%** of respondents added compliant detail on their trip



Stop labels (word cloud, in Dutch)



Stop motives (count)

1

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## EXERCISE

# What sensors do you have on you? And can it be used for research?

1

Together with your neighbor, list every sensor you have on you (e.g. on your phone, watch, earbuds, laptop)

2

Pick one sensor. What could you measure with this? What could go wrong when doing this for research? (e.g. ethically or practically)

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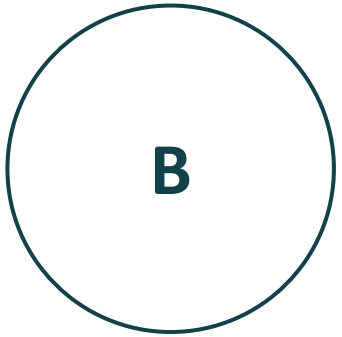
Exercise

Exercise

## Some example answers

*More right answers are possible of course!*

| SENSOR                  | WHAT YOU COULD MEASURE                                        | WHAT COULD GO WRONG                                                        |
|-------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------|
| Phone GPS / Wi-Fi       | Daily mobility, commute patterns, time spent at locations     | Re-identification risk if shared at high spatial or time resolution        |
| Accelerometer / steps   | Physical activity, sleep-wake rhythm                          | Sensitive health inferences, even without asking about health              |
| Microphone (ambient)    | Noise level, environment or context, not conversation content | Still feels invasive; needs clear consent and no raw audio storage         |
| App usage / screen time | Digital wellbeing, attention and routine patterns             | Broad permissions needed; can reveal sensitive apps (health, dating, etc.) |



## Capture and scraping approaches



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# Capture and scraping approaches — definitions

## DEFINITION

### Screen capture

A software application installed on the participant's device that records what's on their screen without relying on the phone's own sensors. This can happen continuously or as periodic screenshots.

## DEFINITION

### Scraping

A software application that automatically collects and extracts specific content (text, images, page structure) from digital sources such as an app or a website. It does not create a full visual copy of the screen.

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Exercise

## How does it work in practice?

1

**Invite**

You invite participants to your study.

2

**Install**

They install an app or plug-in that captures their screen or scrapes specific content.

3

**Collect**

Data collection is prospective: screenshots or scraped content are logged automatically as they use their device.

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## EXAMPLE

## The human screenome project

**Screenomics** software records, encrypts and transmits screenshots automatically and unobtrusively

**every 5 seconds**, whenever a device is turned on.

When deployed on multiple devices at once, the screenshots from each one are synced in time.

|    | Pala Alto Transit Center (Caltrain Platform South, Drop-off) | Pala Alto Transit Center (Caltrain Platform South, Pick-up) | San Jose (Quarry) |
|----|--------------------------------------------------------------|-------------------------------------------------------------|-------------------|
| AM | 12:40 AM                                                     | 12:41 AM                                                    | 12:42             |
| AM | 12:55 AM                                                     | 12:56 AM                                                    | 12:57             |
| AM | 1:10 AM                                                      | 1:11 AM                                                     | 1:12              |
| AM | 1:25 AM                                                      | 1:26 AM                                                     | 1:27              |
| PM | 12:40 PM                                                     | 12:41 PM                                                    | 12:42             |
| PM | 12:55 PM                                                     | 12:56 PM                                                    | 12:57             |
| PM | 1:10 PM                                                      | 1:11 PM                                                     | 1:12              |
| PM | 1:25 PM                                                      | 1:26 PM                                                     | 1:27              |
| PM | 1:40 PM                                                      | 1:41 PM                                                     | 1:42              |
| PM | 1:55 PM                                                      | 1:56 PM                                                     | 1:57              |
| PM | 2:10 PM                                                      | 2:11 PM                                                     | 2:12              |
| PM | 2:25 PM                                                      | 2:26 PM                                                     | 2:27              |
| PM | 2:40 PM                                                      | 2:41 PM                                                     | 2:42              |

|  | Pala Alto Transit Center (Caltrain Platform South) | Campus Drive (Pala, DC) | Pala Alto Transit Center (Caltrain Platform South) |
|--|----------------------------------------------------|-------------------------|----------------------------------------------------|
|  | 1:10 PM                                            | 1:20 PM                 | 1:30 PM                                            |
|  | 1:40 PM                                            | 1:50 PM                 | 2:00 PM                                            |
|  | 2:10 PM                                            | 2:20 PM                 | 2:30 PM                                            |
|  | 2:40 PM                                            | 2:50 PM                 | 3:00 PM                                            |

Screenshots collected for one participant

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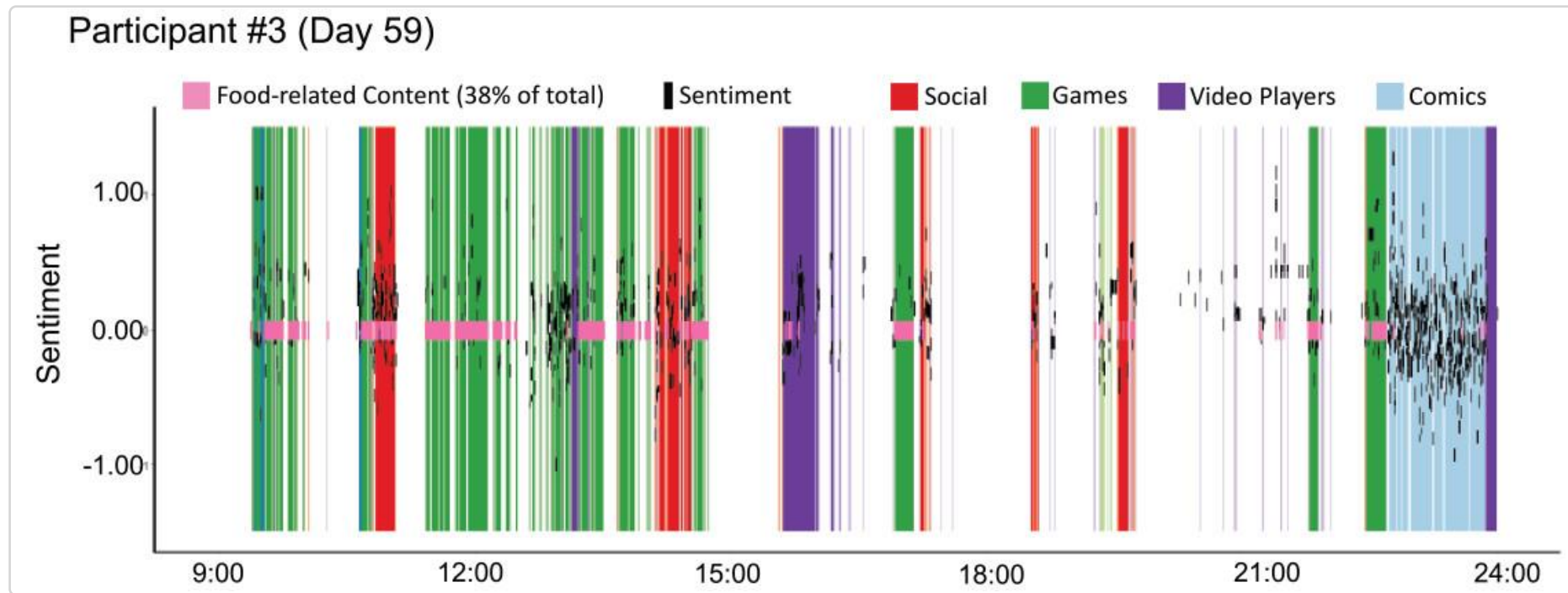
Example

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## EXAMPLE

## Example data from the Screenomics app



Participant #3, Day 59: App category and sentiment reconstructed from screenshots taken every 5 seconds

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## EXERCISE

**Discuss with your neighbor: If a screenshot is made every 5 seconds ...**

1

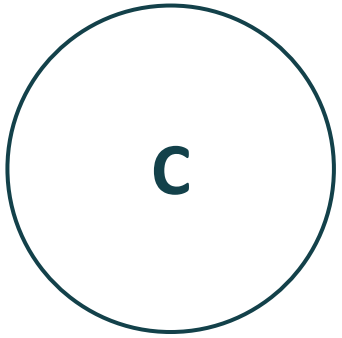
**What would this reveal about you that GPS or step count never could?**

2

**Think about what your Screenome from yesterday would have looked like. What is one thing that would be surprising about this?**

# 15

MINUTE COFFEE BREAK



## **Data donation approaches**

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# GDPR – definitions

## DEFINITION

### Personal data

Information relating to an identified or identifiable natural person

## DEFINITION

### Data subject

The natural person that the personal data refer to.

## DEFINITION

### Data controller

The natural person or organization responsible for processing personal data. The controller decides which data will be processed, how and why.



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# Data donation — definitions

## DEFINITION

### Data Download Package (DDP)

Under GDPR's right of access and data portability, platforms must provide users their personal data in a machine-readable format, typically a .zip file.

## DEFINITION

### Local processing

Extracting only the data that is relevant to a research question directly on the participant's own device, before anything is shared with the researcher.

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## How does it work in practice?

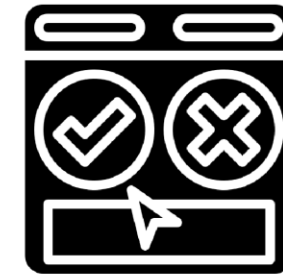
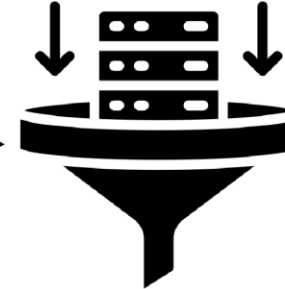
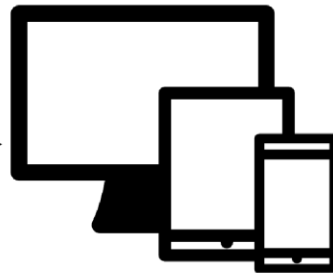
Informed consent

Data Download Packages (DDPs)

Local processing

Inspect and donate

Data analysis



Participant

Digital trace data

Participant's device

Variables of interest

Researcher

1

**Request**

The participant requests their DDP directly from the platform.

2

**Review & select**

They open the package locally and choose what to donate.

3

**Donate**

The selected data is submitted to the researcher; the rest stays on the participant's device.

**Two ways to do data donation:** share the complete DDP, or process it locally and share only what's relevant. Local processing is now the more common approach.

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## EXAMPLE 1

## How common is binge-watching?

### WHAT'S IN THE NETFLIX DDP

| ViewingActivity.csv |                  |       |                       |
|---------------------|------------------|-------|-----------------------|
| Profile             | Start time       | Dur.  | Title                 |
| Alex                | 2024-03-02 21:14 | 42:10 | Stranger Things S3:E4 |
| Alex                | 2024-03-02 21:58 | 38:22 | Stranger Things S3:E5 |
| Alex                | 2024-03-02 22:37 | 40:05 | Stranger Things S3:E6 |
| Jamie               | 2024-03-03 08:12 | 12:44 | Bluey S2:E1           |
| Jamie               | 2024-03-03 19:41 | 51:30 | The Crown S4:E2       |

### AFTER PROCESSING

| MIN. ITEMS IN A SESSION | % BINGE |
|-------------------------|---------|
| Two items               | 32.2%   |
| Three items             | 14.8%   |
| Four items              | 8%      |
| Five items              | 4.7%    |
| Six items               | 2.8%    |

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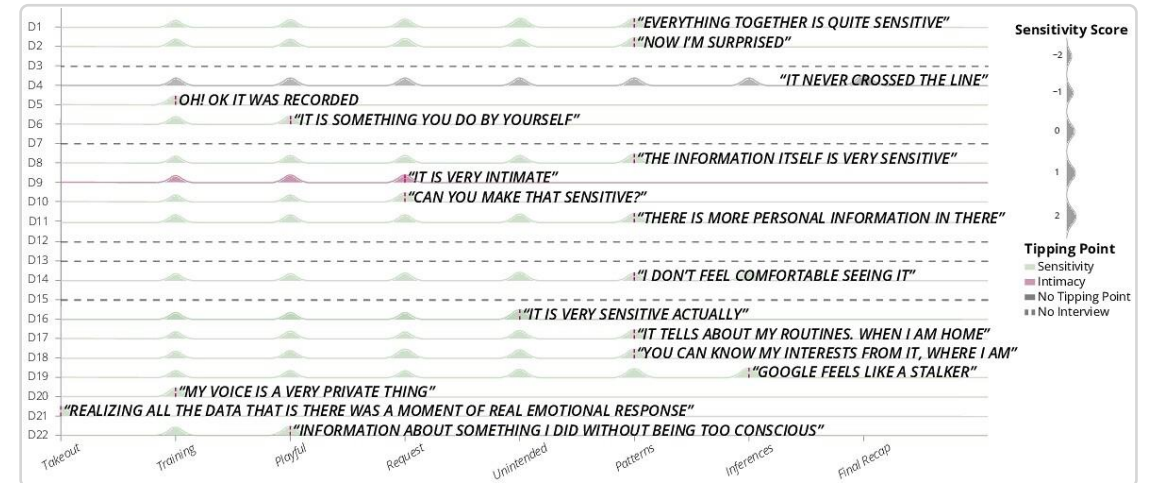
## EXAMPLE 2

## Sensitivity and intimacy with Google Assistant users

## WHAT'S IN THE DDP

| MyActivity.json |                                 |
|-----------------|---------------------------------|
| Time            | Query (verbatim)                |
| 14:02           | "Remind me to call mom at 5"    |
| 14:15           | "What's the weather tomorrow"   |
| 19:41           | "Play calming sleep sounds"     |
| 21:12           | "Am I stressed, how can I tell" |
| 21:13           | "How to stop feeling anxious"   |

## AFTER PROCESSING



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## EXERCISE

**A researcher wants to use TikTok data donation for the RQ: “Does short-form video use before bed predict how well someone sleeps?”**

1

What is in the raw TikTok DDP? (you cannot look it up directly, but you can guess...)

2

How could you process this into measurements that answer the research question?

3

What can be done locally on the participant’s device?

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Exercise

## Suggested answers

TikTok DDP: watch history, search history, videos liked/shared/commented on, login/session activity, direct messages, following/followers, etc.

Process into measurements: take timestamps around “before bedtime” → compute summary statistics (e.g. total watch time, number of videos, etc.); combine with a survey on sleep (quality, time to fall asleep, time to wake up), to correlate the measures with.

What can be done locally: processing the timestamps into summary statistics over the nights.

03

## **Advantages, challenges, ethical considerations**

# Advantages

## All approaches

- Naturalistic context: behavior is observed as it happens, not staged for the study
- Large scale: across many participants or data points
- Detailed, granular data
- Captures concepts that were previously hard to measure
- Can be supplemented with other data sources, e.g. surveys

## Sensor-based

- Relatively low participant effort: Sensors run passively in the background

## Capture / scraping

- Relatively low participant effort: Once installed, the tool runs unobtrusively

## Data donation

- Data is collected retrospectively.
- Not influenced by participants “feeling seen” while it's collected
- Relatively cheap to set up: No custom app or sensor deployment needed (only potentially set-up local processing)

# Challenges

## All approaches

- Not everyone can, or wants to, participate
- Substantial data processing needed before data is usable
- Non-participation due to privacy concerns
- Participants need to have digital skills

## Sensor-based

- App or sensor development is expensive
- Expensive to maintain over time

## Capture / scraping

- Some platforms actively block scraping
- Requires active maintenance as platforms change

## Data donation

- Substantive effort required from participants: Requesting and reviewing their own data
- Every study needs a custom design; no one-size-fits-all pipeline

# Ethical considerations

## All approaches

- Data passes through multiple stages: Carefully consider who has access at each stage

## Sensor-based

- Participants may forget they are being observed over time

## Capture / scraping

- Participants may forget they're being observed (as in screenomics)
- Information of contacts may be revealed incidentally

## Data donation

- Participants can inspect data before sharing → but can they really oversee such large volumes of data?
- Contacts' information may be revealed too. But local processing can help prevent this

## EXERCISE

Pick a method for this research question:

*“How does exposure to news on social media affect people's mood throughout the day?”*

- 1 **Choose a method:** Sensor based, capture/scraping, or data donation?
- 2 **Explain why:** Name one advantage that makes it the right fit, and one challenge or ethical risk you'd need to manage.

## EXERCISE

# Suggested answers

| Method                                                             | Advantage                                                                                                 | Challenge / ethical risk                                                                  |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Data donation of a social media platform together with EMA on mood | It captures actual news exposure (e.g. Which posts, how many, when?) and can be linked to the EMA answers | News exposure is mixed with exposure to all other things. Can it be filtered out locally? |
| Sensor based                                                       | Passively capture exposure                                                                                | Sensors can only see app-level activity (not the content)                                 |
| Screen capture                                                     | Rich collection of all content seen, including all news                                                   | You might also capture a lot of other sensitive information on screen                     |

04

## Found and designed data

# Found data, designed data

From a researcher's perspective, big data sources look **"found"**: already out there, waiting to be used. But someone **"designed"** it first.

## DEFINITION

### Found data

Data that already exists “out there”, which researchers didn't set out to create. Examples are platform logs, transaction records, social media posts. It was already there, before it was collected for a (research) purpose.

## DEFINITION

### Designed data

The same data, seen from another angle: every field, format, and default was chosen by someone (e.g. a platform, a research team) for a reason of their own. This means that no aspect of the data is neutral or automatic.

# Two questions to ask of any data source

1

## Active or passive?

Was this data actively created by the person (e.g. typing a message, writing a post) or passively logged as a by-product of just using a device or platform?

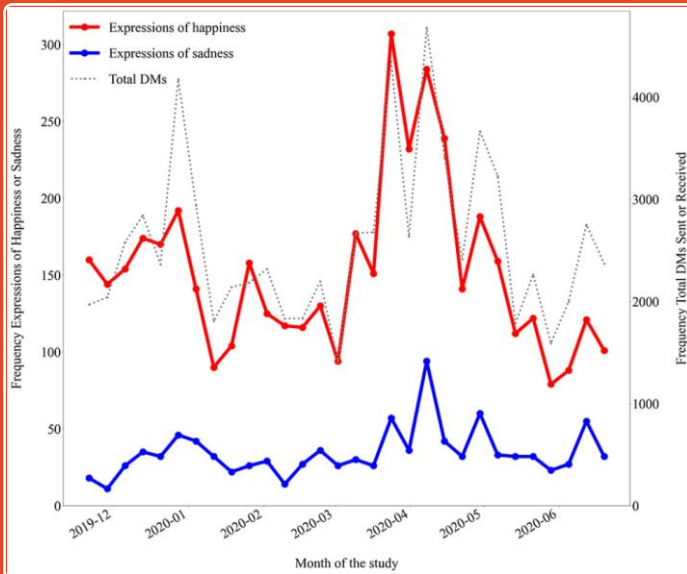
2

## Who designed the data, and why?

Every field that got captured, every option offered, every default that was set has been decided on by someone for their own reasons. What were their incentives to shape the data like this?

## EXERCISE

# Look again at Project AweSome



*Happiness and sadness in adolescents' Instagram DMs*

1

Was this data collection active or passive?

Different aspects: messages, sentiment scores ...

2

Who designed the data, and why?

E.g. think of the DM feature of Instagram.

## EXERCISE

# Suggested answers

*Happiness and sadness in adolescents' Instagram DMs (Verbeij et al., 2024)*

### Active or passive?

The DMs themselves are actively written by the teens.

Sentiment scores are passively derived: a dictionary or classifier scores the text after the fact. No participant actively reported their own mood.

### Who designed the data, and why?

Instagram designed the DM feature to maximise engagement and time-in-app, including message length, timestamps, whether read-receipts exists → product decisions.

### Why is the difference important?

Sentiment tools are often built and validated on adult or generic text. Teen slang, sarcasm, and abbreviations can be misread by the classifier. As a result, “found” data quality depends heavily on how well it fits a purpose it was never built for.



# Questions?

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